

SSD_Default ENV Setup Guide

If the ENV partition is suddenly powered off during the reading and writing process, there is a very small probability that data will be damaged, resulting in the loss of ENV. Once the data is lost, the system will fail to start. In order to ensure the normal startup of the system, when the ENV partition data CRC check fails, it will automatically start from the default value. Since the settings for each user partition are different, follow the two steps below to configure the defaults where appropriate.

The following log represents ENV CRC check failure:

```
*** warning - bad CRC, using default environment
```

Here is an example of NAND flash (*NOR flash can jump directly to the second step)

1. Set default mtdpart: (partition information)

`boot/include/configs/infinity2m.h` The corresponding macro definition MTDPARTS_DEFAULT in the file needs to be modified .

Note: The mtdpart here is the same as the one after entering mtdpart under uboot.

Eg: Enter mtdpart under uboot


```
AL - [infinity2m.h (Y:\alkaid\boot\include\configs) *]
pls View Window Help
on) environment.h (include) env_common.c (common) env_flash.c (common) env_nand.c (common) env_sf.c (common) fw_
400: /* SENSOR */
401: #define CONFIG_MS_SRCFG
402:
403: /* default env */
404:
405: #define CONFIG_BOOTARGS "ttyS0,115200 ubi.mtd=UBI,2048 root=ubi:rootfs ro ro
406:
407: #define CONFIG_BOOTCOMMAND
408: "bootlogo 0 0 0 0 0; "
409: "mw 1f001cc0 11; "
410: "gpio out 8 0; "
411: "nand read.e 0x22000000 KERNEL 0x500000; "
412: "gpio out 8 1; "
413: "bootm 0x22000000; "
414: "nand read.e 0x22000000 RECOVERY 0x500000; "
415: "bootm 0x22000000;"
416:
```

3. Set custom Default ENV in Uboot

Modify the file: `boot\include\env_default.h`

Add default ENVs in the `default_environment[]` array with reference to other ENVs (*bootargs and bootcmd set above will eventually be defined in this array).

```
19: {
20: #elif defined(DEFAULT_ENV_INSTANCE_STATIC)
21: static char default_environment[] = {
22: #else
23: const uchar default_environment[] = {
24: #endif
25: "my_env=" "my_env_test" "\0"
26:
27: #ifdef CONFIG_ENV_CALLBACK_LIST_DEFAULT
28: ENV_CALLBACK_VAR "=" CONFIG_ENV_CALLBACK_LIST_DEFAULT
29: #endif
30: #ifdef CONFIG_ENV_FLAGS_LIST_DEFAULT
31: ENV_FLAGS_VAR "=" CONFIG_ENV_FLAGS_LIST_DEFAULT "\0"
```

```
SigmaStar # printenv my_env
my_env=my_env_test
```

After `boot/include/configs/infinity2m.h` setting the default env necessary for system startup, recompile uboot and update it to the board to ensure that the system can start normally when the env data is lost.